Benchmarking Framework for Large-Scale Modular Quantum Systems

Technical Research Paper v2.1

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Abstract

This paper defines the benchmarking framework that QONTOS should use to measure progress toward large-scale modular quantum computing. It does not present million-qubit benchmark values as measured QONTOS results. Instead, it establishes benchmark categories, measurement conditions, scenario-based targets, and validation gates spanning simulator-backed orchestration, digital-twin modeling, hardware integration, and eventual fault-tolerant modular systems. The framework includes system metrics, logical-qubit metrics, and application-level benchmarks, with FeMoco retained as a flagship stretch benchmark rather than the sole validation target.

Claim status: methodology paper for benchmark design and release gating

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1. Why Benchmarking Must Lead the Narrative

The QONTOS research program will only remain credible if every major architecture and roadmap claim can be mapped to a benchmark, protocol, or validation gate.

This paper therefore serves a special role:

- it defines how progress will be measured
- it distinguishes measured values from targets
- it tells readers how to compare current QONTOS capability with future scenarios

1.1 Claim Status Summary

Claim	Status
QONTOS has a current simulator-backed benchmark capability	Supported internally
QONTOS has measured million-qubit benchmark results	Not supported
QONTOS has stretch benchmark targets for million-qubit systems	QONTOS target / stretch target

1.2 Benchmark Categories

QONTOS should report benchmarks in four categories:

Category	Meaning
Implemented platform	Measured on current software stack and benchmark harness
Digital twin	Simulated in modular hardware and runtime models
Hardware integration	Measured on real devices or module prototypes
Stretch architecture	Scenario target under explicit assumptions

2. Measurement Conditions

Every reported benchmark should identify:

- claim label
- environment
- workload
- noise model
- hardware assumptions
- reconstruction or approximation method
- confidence interval or uncertainty note

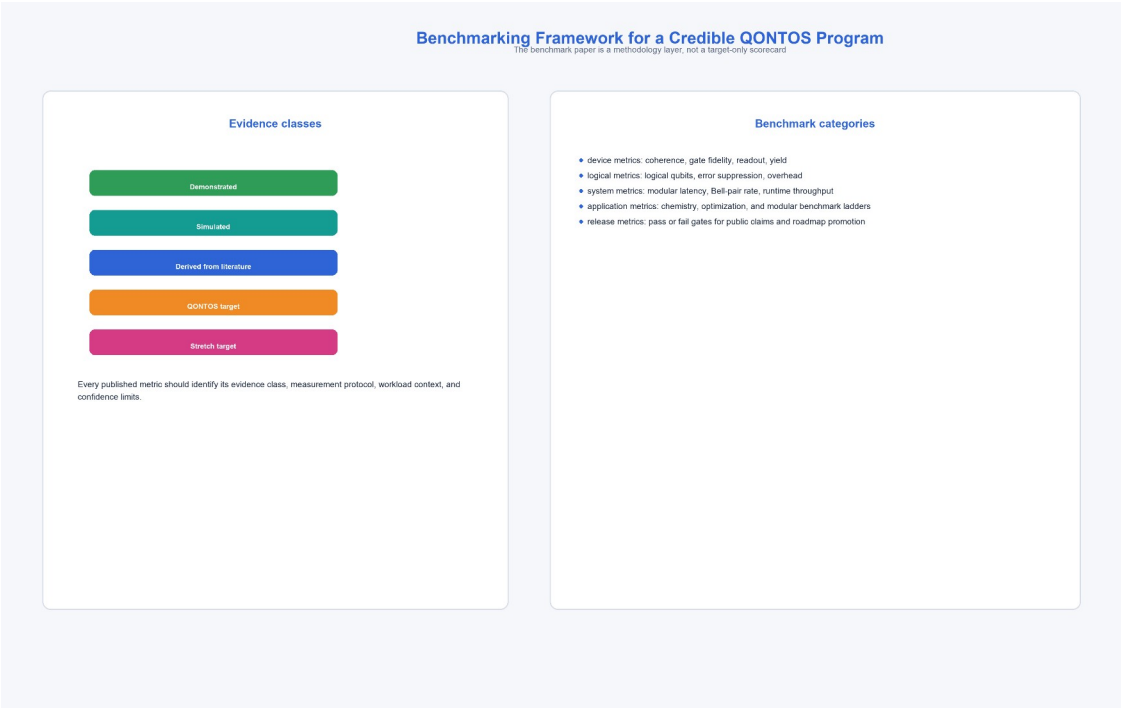
2.1 Mandatory Metadata for Every Result

Field
claim label
benchmark layer
simulator, digital twin, or hardware
workload size
noise or fidelity assumptions
partitioning / modular assumptions
pass/fail threshold

2.2 Benchmark Layers

Layer
L0
L1
L2
L3

3. System Benchmarks



Benchmarking Framework

Figure 1: QONTOS benchmarking should classify every metric by evidence level and benchmark type.

3.1 Metric Definitions

Metric	What it means	Safe use
Quantum Volume	system-level randomized benchmark	only if measurement protocol is explicit
CLOPS	circuit layer throughput	only if execution environment is explicit
Logical qubit count	simultaneously usable logical resources	must define what counts as logical
Logical error rate	post-correction error performance	must include code and decoder assumptions

3.2 Scenario Targets

Metric	Base	Aggressive	Stretch	Claim posture
Logical qubits	1-20	100-1,000	10,000	target range
Quantum	benchmark-	benchmark-	10^15	stretch

Volume	defined	defined		target
CLOPS	benchmark-defined	benchmark-defined	10^12	stretch target
Runtime-supported circuit depth	benchmark-defined	benchmark-defined	10^12-scale flagship class	stretch target

3.3 Competitive Comparison Rule

Do not mix:

- measured IBM or Google values
- QONTOS stretch targets

inside the same table without clearly separating:

- measured current values
- literature projections
- QONTOS targets

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4. Logical-Qubit Metrics

4.1 Required Definitions

QONTOS should define a logical qubit as usable only if:

- the code family is specified
- the decoder path is specified
- the correction cycle budget is specified
- the logical error target is specified

4.2 Scenario-Based Logical Metrics

Metric	Base	Aggressive	Stretch
Effective overhead	1000:1	300:1	100:1
Logical error target	1e-4	1e-6	1e-8
Logical coherence goal	benchmark-defined	benchmark-defined	long-duration flagship workloads

4.3 What Not To Do

Do not report:

- logical qubit count
- logical coherence

- logical advantage

without stating the exact error-correction, decoder, and architecture assumptions.

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5. Application Benchmark Ladder

QONTOS should benchmark progress using a staircase of workloads rather than jumping directly to FeMoco.

5.1 Chemistry Ladder

Stage
C1
C2
C3
C4

5.2 Optimization Ladder

Stage
O1
O2
O3

5.3 Flagship Benchmark Rule

FeMoco should remain the flagship stretch benchmark, but it must not be the only benchmark.

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6. Current QONTOS Benchmark Surface

The current QONTOS benchmark story should stay tied to the implemented software-first platform:

- simulator-backed orchestration
- partitioning and scheduling experiments
- digital-twin projections
- result aggregation and replayability

This is the safe foundation from which future modular and hardware benchmarks can grow.

6.1 Current Safe Public Framing

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QONTOS has a benchmark program for modular orchestration and digital-
twin validation today, and is defining a larger benchmark framework
for future hardware-integrated modular systems.
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7. Pass/Fail Gates

Each benchmark family should have an explicit gate.

Gate	Required outcome
G1	benchmark protocol defined and reproducible
G2	current platform benchmarks run cleanly and are versioned
G3	digital-twin results can be reproduced from assumptions
G4	hardware results, when present, are clearly separated from targets
G5	flagship application claims are tied to a benchmark ladder, not a single heroic endpoint

8. Release and Communication Rules

8.1 Safe benchmark wording

- target
- stretch benchmark
- digital-twin projection
- simulator-backed result
- measured on current platform

8.2 Unsafe benchmark wording unless measured

- QONTOS achieves Quantum Volume 10^15
- QONTOS achieves CLOPS 10^12
- QONTOS has 10,000 logical qubits operational
- QONTOS demonstrates quantum advantage

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9. Conclusion

Benchmarking should be the credibility engine of the QONTOS research and production program. The most important shift in this paper is from target declaration to measurement design. Once QONTOS can show how it defines, measures, and gates each major claim, the rest of the paper set becomes much more technically defensible.

References

[1] Representative literature on Quantum Volume, CLOPS, logical benchmarking, and application-level FTQC evaluation.

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